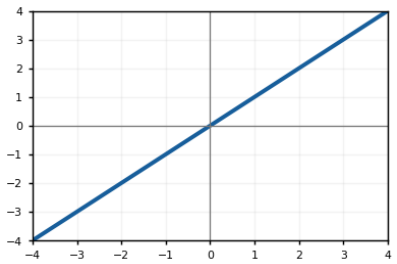
The 12 Basic Functions

Part A | Graph gallery — all 12 graphs on one page

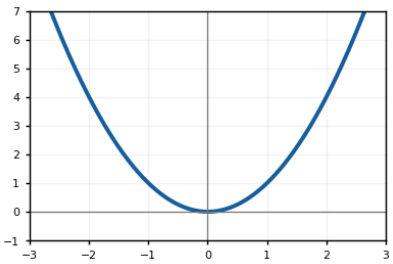
Match graphs A–L to the function names and rules on page 2.

Graphs are schematic and are not necessarily drawn to the same scale.

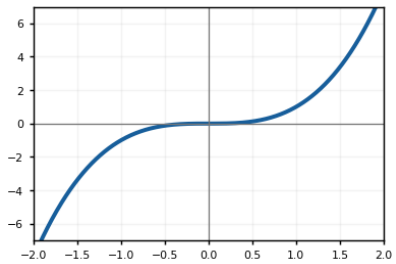
Graph A



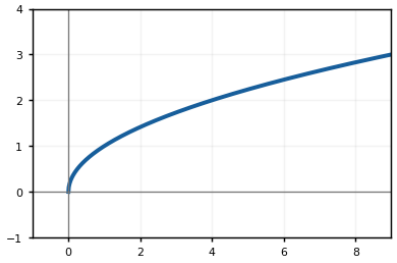
Graph B



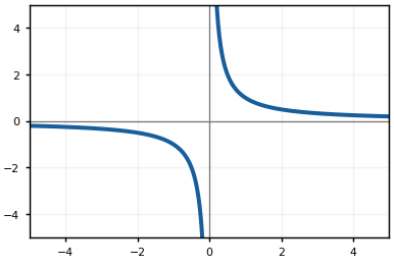
Graph C



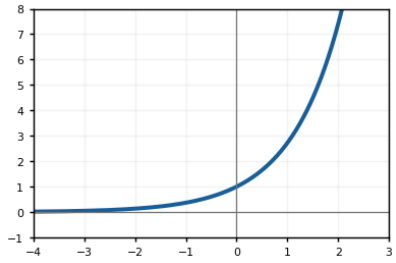
Graph D



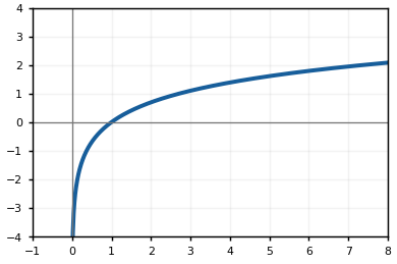
Graph E



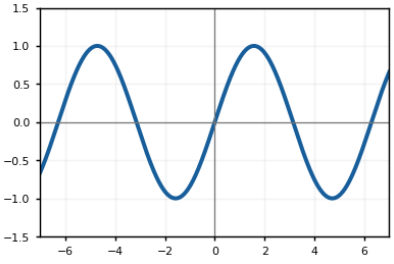
Graph F



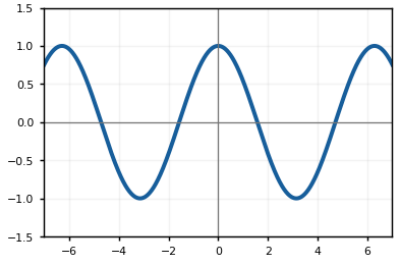
Graph G



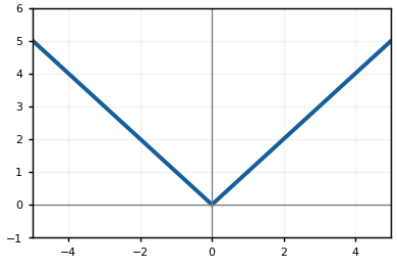
Graph H



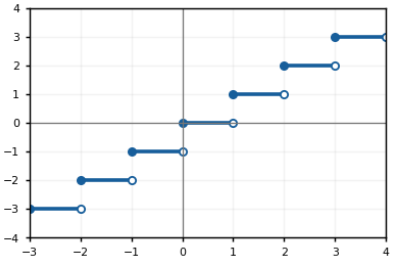
Graph I



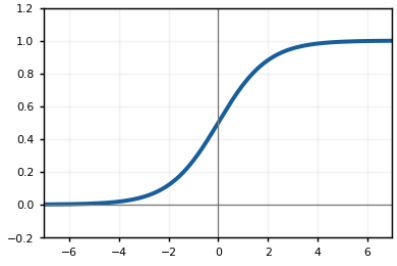
Graph J



Graph K



Graph L



Part A | Matching answer sheet

Write the graph letter beside each function name and rule.

GRAPH	FUNCTION NAME	FUNCTION RULE
_____	Identity	$y = x$
_____	Squaring	$y = x^2$
_____	Cubing	$y = x^3$
_____	Square root	$y = \sqrt{x}$
_____	Reciprocal	$y = \frac{1}{x}$
_____	Exponential	$y = e^x$
_____	Natural logarithm	$y = \ln(x)$
_____	Sine	$y = \sin(x)$
_____	Cosine	$y = \cos(x)$
_____	Absolute value	$y = \lvert x \rvert$
_____	Greatest integer (floor)	$y = \lfloor x \rfloor$
_____	Logistic	$y = \frac{1}{1+e^{-x}}$

Remember: domain = allowed x-values; range = resulting y-values.

Use [ ] for included endpoints, ( ) for excluded endpoints and infinity.

Part B | Multiple-choice practice

Questions 1–12 • Circle one answer per question

1. What is the domain of the square-root function? $y = \sqrt{x}$

A. $(-\infty, \infty)$ B. $[0, \infty)$

C. $(0, \infty)$ D. $(-\infty, 0]$

2. What is the range of the square-root function? $y = \sqrt{x}$

A. $[0, \infty)$ B. $(-\infty, \infty)$

C. $(0, \infty)$ D. $[-1, 1]$

3. What is the domain of the reciprocal function? $y = \frac{1}{x}$

A. $[0, \infty)$ B. $(-\infty, 0]$

C. $(-\infty, 0) \cup (0, \infty)$ D. $(-\infty, \infty)$

4. What is the range of the reciprocal function? $y = \frac{1}{x}$

A. $(-\infty, \infty)$ B. $[0, \infty)$

C. $(-\infty, 0) \cup (0, \infty)$ D. $(0, \infty)$

5. Which function has a minimum at (0, 0) and no maximum?

A. Cubing B. Squaring

C. Identity D. Reciprocal

6. On which interval is the squaring function decreasing?

A. $(0, \infty)$ B. $(-\infty, 0)$

C. $(-\infty, \infty)$ D. Never

7. Which function is increasing over its entire domain?

A. Cubing B. Absolute value

C. Squaring D. Cosine

8. What are the x- and y-intercepts of the absolute value function?

A. (1, 0) only B. (0, 1) only

C. (0, 0) for both D. None

9. What is the y-intercept of the exponential function? $y = e^x$

A. (0, 0) B. (0, 1)

C. (1, 0) D. None

10. Which statement about the exponential function is true? $y = e^x$

A. Range is $(-\infty, \infty)$ B. It crosses the x-axis

C. Horizontal asymptote $y = 0$ D. It decreases

11. What is the x-intercept of the natural logarithm? $y = \ln(x)$

A. (0, 0) B. (1, 0)

C. (0, 1) D. None

12. What is the domain of the natural logarithm? $y = \ln(x)$

A. $[0, \infty)$ B. $(-\infty, \infty)$

C. $(0, \infty)$ D. $(-\infty, 0)$

Part B | Multiple-choice practice

Questions 13–24 • Circle one answer per question

13. What is the range of the sine function?

A. $(-1, 1)$
B. $[-1, 1]$
C. $[0, 1]$
D. $(-\infty, \infty)$

14. What is the maximum value of the cosine function?

A. 0
B. 2
C. 1
D. No maximum

15. Which basic function has y-intercept (0, 1) and oscillates?

A. Sine
B. Cosine
C. Natural logarithm
D. Cubing

16. On which interval is absolute value decreasing?

A. $(-\infty, 0)$
B. $(0, \infty)$
C. $(-\infty, \infty)$
D. Never

17. What is the range of the greatest-integer function?

A. $[0, \infty)$
B. All real numbers
C. All integers
D. $(-1, 1)$

18. What is the value of the greatest integer less than or equal to −1.3?

A. -1
B. -2
C. 1
D. 2

19. What is the range of the logistic function? $y = \frac{1}{1+e^{-x}}$

A. $[0, 1]$
B. $(0, 1)$
C. $(-\infty, \infty)$
D. $[-1, 1]$

20. What is the y-intercept of the logistic function?

A. (0, 0)
B. (0, 1)
C. (0, ½)
D. None

21. Which function has asymptotes x = 0 and y = 0?

A. Reciprocal
B. Natural logarithm
C. Exponential
D. Square root

22. Which statement about the cubing function is correct?

A. Bounded above only
B. Bounded below only
C. Bounded above and below
D. Unbounded above and below

23. Which interval is the sine function increasing on?

A. $(-\pi/2, \pi/2)$
B. $(0, \pi)$
C. $(\pi/2, 3\pi/2)$
D. $(-\pi, 0)$

24. Which function has no x-intercept and is bounded above and below?

A. Squaring
B. Exponential
C. Logistic
D. Identity

Teacher answer key

Keep this page separate when distributing the student assignment.

PART A | Graph matches

Identity: K

Squaring: C

Cubing: H

Square root: F

Reciprocal: B

Exponential: E

Natural logarithm: I

Sine: L

Cosine: A

Absolute value: J

Greatest integer (floor): D

Logistic: G

PART B | Multiple-choice answers

1. B	2. A	3. C	4. C	5. B	6. B
7. A	8. C	9. B	10. C	11. B	12. C
13. B	14. C	15. B	16. A	17. C	18. B
19. B	20. C	21. A	22. D	23. A	24. C

Teaching notes

- Intervals of increase/decrease use open endpoints at turning points.
- The floor function has a jump at each integer.
- The logistic function approaches 0 and 1 but never reaches either value.
- For sine and cosine, intervals repeat periodically.